

DESIGN AND ANALYSIS OF ALGORITHMS

Practical No. 4: Strassen's Matrix Multiplication

1. Aim

To multiply two square matrices using Strassen's Matrix Multiplication algorithm and study its efficiency.

2. Objective

- Understand divide-and-conquer matrix multiplication.
- Reduce the number of recursive multiplications.
- Compare Strassen's method with conventional multiplication.

3. Theory

Conventional multiplication of two $n \times n$ matrices requires $O(n^3)$ scalar multiplications. Strassen's algorithm divides each matrix into four submatrices and uses seven recursive multiplications instead of eight, together with additions and subtractions. Its asymptotic time complexity is $O(n^{\log_2 7})$, approximately $O(n^{2.807})$, which is better than $O(n^3)$ for sufficiently large matrices.

4. Conclusion

Strassen's algorithm performs matrix multiplication with fewer recursive multiplications and demonstrates how divide and conquer can improve asymptotic efficiency.